



**UNSW  
COMM5000  
T3 2026**

# **Course and Milestone 1 project**

**Used Cars**

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## Used Cars project summary

India's used car market has grown rapidly in recent years and now sells more vehicles each year than the new car market. As a result, online marketplaces and dealer groups are investing in new technologies to enhance both vehicle pricing and sales. Accurate price appraisal plays a vital role in these processes and currently relies heavily on manual valuations by dealers and inspection experts.

You consult for an online used car marketplace operating across a number of major Indian cities, and you help this company to predict or estimate the market price of a used car at the point of appraisal.

Knowing the expected price will allow the marketplace to be better positioned to make instant cash offers, flag under- and over-priced listings, and predict inventory turnover and yearly sales. It will also support the appraisal teams by potentially improving the quality and speed of their decisions. Furthermore, similar techniques can help in target marketing by modelling buyer preferences in niche market segments.

In order to predict used car prices you will use a dataset consisting of 1,005,000 used car listings covering 13 brands across a number of Indian cities, and the statistical methods covered in this course.

## IN THIS PROJECT (M1, M2, and final report) YOU WILL:



As a consultant, do a preliminary Exploratory Data Analysis over the dataset.

The results from your M1 may be used for the analysis in M2 and in the final report.



Synthesise your potential insights from the M1 and M2 and construct a model using exclusively and only the variables provided in the dataset. More on this in the final report.



Based on your analysis, deliver your findings to the company you are consulting.

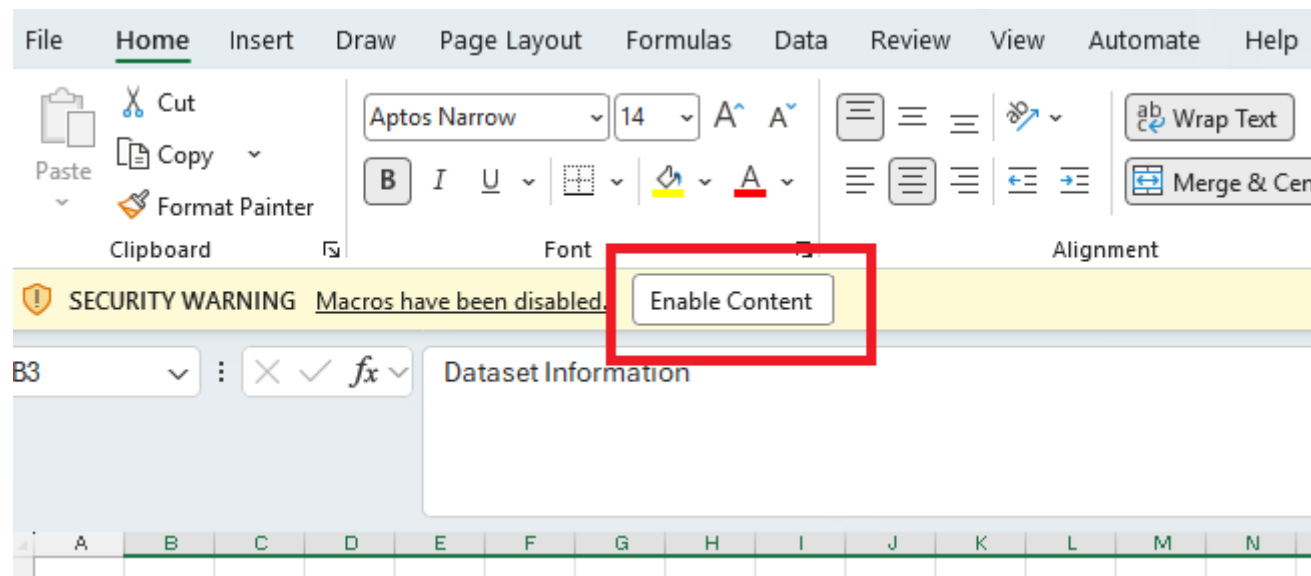
## THE DATA

**Dataset file:** This assessment requires the download of the Excel file provided on our course Moodle page

The file name is: "*COMM5000-Used\_Car\_Price\_Prediction.xlsx*").

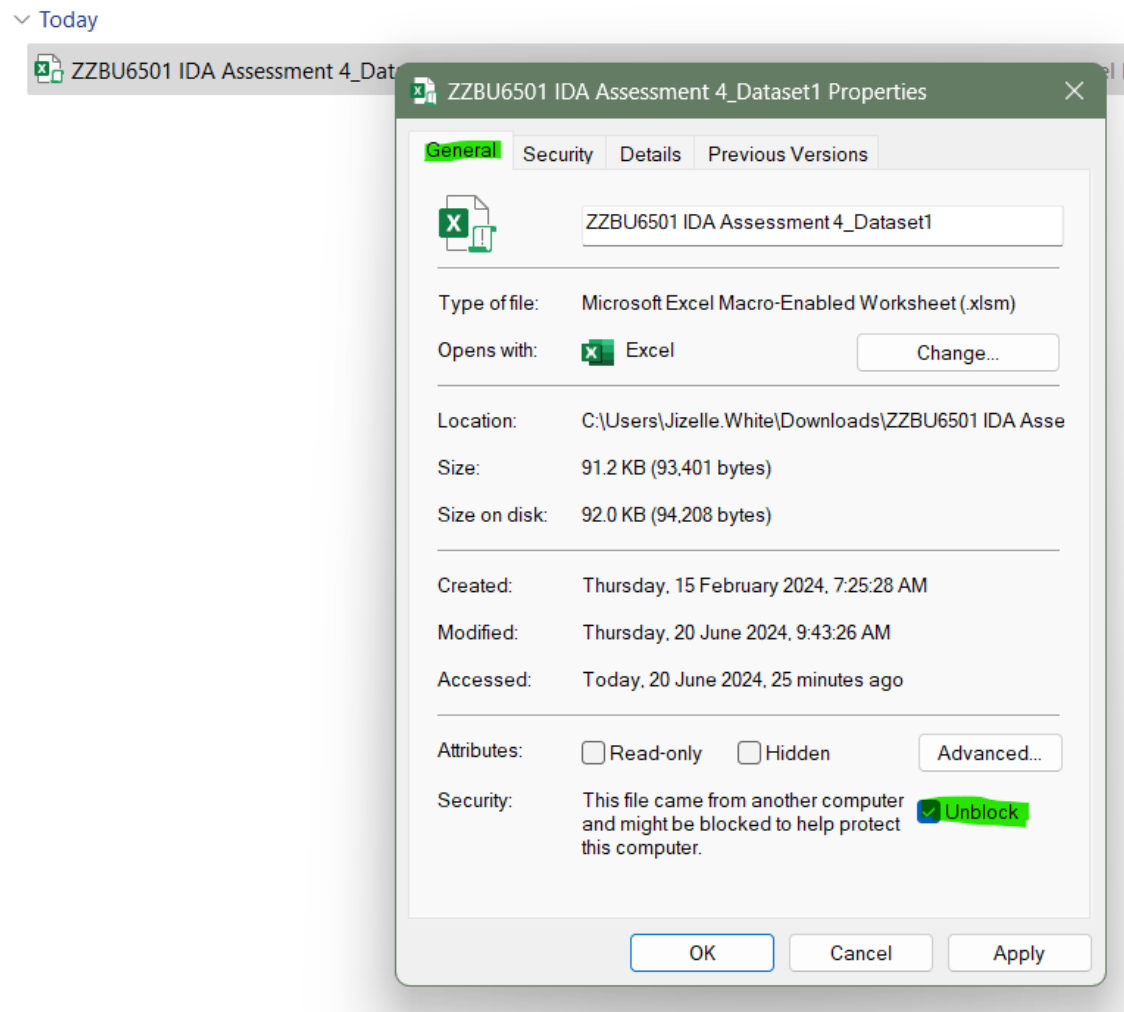
# WHAT IS NEXT?

- For many of you, clicking on “**Enable Content**” upon first opening of the file, is all you will need to do.



# WHAT IS NEXT?

- If “Enable Content” is not an option/does not work: **BEFORE** opening the file, **right-click on the Excel file** and select **Properties - General** tab. Under security, ensure **Unblock** is checked and click **Apply**.



# WHAT IS NEXT?

- Depending on your computer and software setup, this step may be required as [Office blocks macros code by default](#). These specific Excel file datasets require macros to be enabled to function correctly and allow for data randomisation. For consistent file use, it is advised that you:
  1. **Download file** ” *COMM5000-Used\_Car\_Price\_Prediction.xlsm*” from our Moodle site
  2. **Unblock the macro via Properties – General** (see previous slide)
  3. **Open and save** the downloaded dataset to a known location, so you can continue to work from the same file
    1. When you save the file, **keep the same XLSM extension**
  4. **Only use the file you have opened and saved at step 1.**

**Do not** download your Excel dataset file multiple times from Moodle, as this will result in you working with different numbers.

# WHAT IS NEXT?

- If the randomization is successful, cell B6 in tab “Dataset” of the Excel file should display a message similar to the below, and you can save this Excel file in any location on your computer from which you like working. This saved version will be the file you will be working from now onwards until the end of the course

The screenshot shows an Excel spreadsheet with the following content:

|   | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | AA | AB | AC | AD |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|----|----|----|----|
| 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |
| 2 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |
| 3 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |
| 4 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |
| 5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |
| 6 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |
| 7 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |
| 8 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |

**Dataset Information**

This 2026 large-scale dataset contains information about over one million observations and 20 features describing various aspects of used cars in India (Stable Space, 2026). The original dataset was published at the following link: <https://www.kaggle.com/datasets/sharmajicoder/used-car-price-prediction-dataset>. The provided data are the only ones to be used. The original dataset contains realistic imperfections listed under: <https://datasetsearch.research.google.com/search?src=2&query=Used%20Car%20Price%20Prediction%20Dataset&docid=L2cvMTFucWZmMDd5eg%3D%3D> (Updated Jun 13, 2026). Missing values are addressed by the randomising script or via inclusion of "Unknown" labels.

**Input variables:**

- 1.Brand: Manufacturer of the vehicle
- 2.Model: Model name
- 3.Year: Manufacturing year
- 4.Mileage\_kmpl: Fuel efficiency (km/l)
- 5.Engine\_CC: Engine displacement in cubic centimeters
- 6.Horsepower: Engine power output
- 7.Fuel\_Type: Fuel category
- 8.Transmission: Manual or Automatic
- 9.Owner\_Type: Ownership history
- 10.Color: Vehicle color
- 11.City: Registration city
- 12.Kms\_Driven: Total distance traveled
- 13.Insurance\_Valid: Insurance status
- 14.Service\_History: Availability of service records
- 15.Accidents: Number of reported accidents
- 16.Tax\_Paid: Road tax payment status
- 17.Number\_of\_Doors: Vehicle door count
- 18.Seats: Seating capacity
- 19.Registration\_Age: Vehicle age since registration

**Output variable: Car\_Price: Target variable, in Rupees**

Data randomisation was applied for university assessment purposes on 2026-09-07 15:44:07. The randomised data in this Excel file likely differs from the original. DO NOT DELETE THIS TEXT.

RECAP: Stable Space (2026). Used Car Price Prediction Dataset (Dataset). <https://www.kaggle.com/datasets/sharmajicoder/used-car-price-prediction-dataset>. Last accessed: June 2026. MIT Licence

- If cell B6 remains empty, then the randomization has NOT been applied. You are to never change the contents of cell B6.

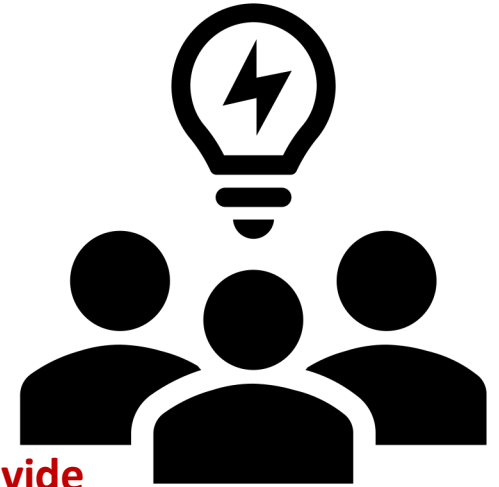
# WHAT IS NEXT?

- Save the spreadsheets to your own device where you will use Excel
- Note: This is an individual work, and worth 20%. You will have to submit your OWN work on the due date
- **Cheating and any other academic integrity violation are not tolerated. We enforce a strict zero-tolerance policy, and any breach could result in severe consequences.**



## M1 Activities

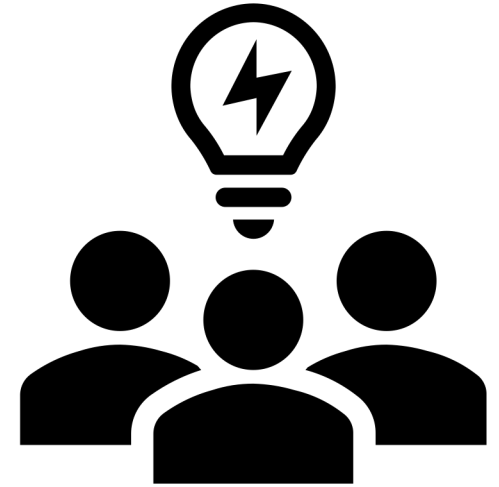
- If you do not upload your report in time, you will not be able to provide feedback on the M1 reports, and you will not be able to participate in the Feedback on feedback activity.
- You will be grouped into teams of 4/5
- The teams will be used to setup peer marking of the individual M1 submission (feedback on feedback: 5%)
- Important to leave high-quality, constructive feedback so that your effort can be rightfully rewarded
- Your feedback will be assessed and you will also have the opportunity to evaluate the feedback you have received.





## M1 Activities

- Timelines:
  - **Week 4**, Friday, deadline for submission of M1
  - **Week 6**, deadline for marking the submissions you were assigned to
  - **Week 7**, deadline for evaluating the feedback you have received
- **20% for M1 report:**
  - Important to submit on time
  - Time allocation for ELS students
- **5% for the quality of feedback provided** (feedback on feedback)
  - An opportunity to learn from each other, and provide **honest, constructive feedback**
  - Computed as **average** of all feedback on feedback received
  - **Teaching team can intervene to address any issue**



# Questions?

- 1) Check discussion forums
- 2) Post on discussion forums
- 3) Email your lecturer or tutor